Nitish Joshi

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Research Interests: Natural Language Processing, Machine Learning

EDUCATION

Indian Institute of Technology, Bombay *2016-2020*
B.Tech in Computer Science & Engineering with *Honors*
GPA: 9.42/10 [1]
Thesis : Learning with Noisy Sequence Labels (supervised by Prof. Preethi Jyothi)

PUBLICATIONS

1. Explore, Propose and Assemble: An Interpretable Model for Multi-Hop Reading Comprehension
Yichen Jiang*, **Nitish Joshi***, Yen-Chun Chen and Mohit Bansal
Proceedings of ACL 2019, Florence, Italy (long paper, oral presentation) [paper]

2. Cross-Lingual Training for Automatic Question Generation
Vishwajeet Kumar, **Nitish Joshi**, Arijit Mukherjee, Ganesh Ramakrishnan and Preethi Jyothi
Proceedings of ACL 2019, Florence, Italy (long paper, oral presentation) [paper]

3. Coupled Training of Sequence-to-Sequence Models for Accented Speech Recognition
Vinit Unni*, **Nitish Joshi*** and Preethi Jyothi
Submitted to ICASSP 2020

* joint first authors

RESEARCH EXPERIENCE

Learning with Noisy Sequence Labels | Undergraduate Thesis July 2019 - Present
Advisor: Prof. Preethi Jyothi, IIT Bombay

- Working on developing unsupervised methods to align and aggregate multiple noisy sequence labels.
- Extending generalization bounds with noisy labels from multi-class classification to sequence tasks.
- Introduced a novel vector quantization based objective to aggregate the noisy labels.

Multi-hop Reading Comprehension [paper] [code] May 2018 - Dec 2018
Advisor: Prof. Mohit Bansal, University of North Carolina, Chapel Hill

- Proposed a novel model which searched and assembled important contextual information, by constructing a tree of reasoning chains to answer a given question for multi-hop QA.
- Conducted extensive ablations and reasoning chain recovery tests to judge the interpretability of model.
- Our model achieved state-of-the-art results on the challenging WikiHop and MedHop datasets.
- Published the work at ACL 2019 as a long paper and publicly released the code for the project.

Cross-lingual Question Generation [paper] July 2018 - Dec 2018
Advisor: Prof. Ganesh Ramakrishnan & Prof. Preethi Jyothi, IIT Bombay

- Worked on building novel models for automatic question generation for low resource languages.
- Proposed cross-lingual unsupervised pre-training techniques which could leverage information from data rich languages like English to improve performance on the downstream task in Hindi and Chinese.
- Obtained improvements on both automatic metrics (like BLEU) and qualitative linguistic evaluation.
- Publicly released a Hindi question answering dataset (HiQUAD); long paper published at ACL 2019.

[1] Including Fall '19

Explainable Natural Language Inference May 2019 - July 2019
Advisor: Dr. Christopher Malon, NEC Labs, Princeton

· Built a multi-hop natural language inference dataset by transforming existing multi-hop QA datasets.
· Proposed models which could generate explanations for the inferential relationship (entailment, neutral or contradiction) without any direct supervision.

Policy Iteration Lower Bounds for Multi-Action MDPs [report] July 2019 - Nov 2019
Advisor: Prof. Shivaram K, IIT Bombay (Course Project)

· Generalized the MDPs proposed in Melekopoglou (1994) which gave exponential lower bound for policy iteration on 2-action MDPs to k actions giving a lower bound of $O(k2^n)$ on simple policy iteration.

Accent Adaptation in Speech Recognition [report] Dec 2017 - May 2018
Advisor: Prof. Preethi Jyothi, IIT Bombay

· Constructed an end-to-end speech recognition system which can learn domain invariant features.
· Incorporated ideas from the field of Domain Adaptation in Computer Vision and NLP such as Domain Adversarial Training of Neural Networks and Multinomial Adversarial Networks.
· Explored Attention-based models and CTC models in automatic speech recognition systems.

SCHOLASTIC ACHIEVEMENTS

- Received Institute Academic Award for academic excellence at IIT Bombay for 2016-17.
- Awarded AP Grade (About top 1% of class) in Linear Algebra and Chemistry Lab.
- All India Rank of 176 in JEE-Advanced among 150,000 shortlisted candidates in 2016.
- Received Silver medal (given to about top 25%) at the USA Physics Olympiad in 2016.
- Placed 8th in State in Regional Mathematics Olympiad, and secured merit certificate in Indian National Mathematics Olympiad in 2014.
- National top 1% in NSEP (Physics Olympiad) and NSEC (Chemistry Olympiad) in India.

PRESENTATIONS & TALKS

- Oral presentation for two long papers on Multi-hop QA and Question Generation at ACL 2019.
- Talk on Explainable Natural Language Inference to the ML Department at NEC Labs, Princeton.
- Presented the paper 'decaNLP: Multitask Learning as QA' as part of the UNC-NLP reading group.

OTHER SELECTED PROJECTS

Self-supervised Learning of Speech Representations [report] July 2019 - Nov 2019
Course Project: Automatic Speech Recognition (ASR)

· Introduced self-supervised tasks to capture contextual (masked prediction) and temporal (sorting) information; Obtained strong empirical results on Speaker Identification and ASR downstream tasks.

Exploiting Correlation in Multi-armed Bandits [slides] July 2019 - Nov 2019
Course Project: Advanced Concentration Inequalities

· Proposed a variant of the C-UCB algorithm. Demonstrated empirical improvements in regret compared to C-UCB and theoretically proved that the algorithm achieves same order-wise regret bounds.

Conditional InfoGAN [report] [code] Jan 2019 - Apr 2019
Course Project: Advanced Machine Learning

· Modified Recurrent Conditional GAN to incorporate the mutual information loss for sequential data.
· Network was able to learn disentangled features for various observable patterns in the output sequence.

Stochastic Algorithms for PCA [report] Jan 2019 - Feb 2019
Self Project

· Reviewed literature on stochastic approximation algorithms for optimization of PCA objective.
· Presented empirical comparison of Oja's Algorithm, Incremental PCA and Matrix Stochastic Gradient.

Parallelizing Query Optimization for Joins [code] July 2018 - Dec 2018
Course Project: Database and Information System

· Implemented parallel algorithms for query optimization of left-deep and bushy tree joins.
· Modified the PostgreSQL back-end code to incorporate the dynamic programming based algorithm.

BiSym - Biological System Simulator [report] [code] Jan 2018 - April 2018
Course Project: Abstractions and Paradigms in Programming

· Designed and implemented a biological cell system using functional programming in Racket.
· Simulated the theory of evolution using various abstractions and higher order functions.

RELEVANT COURSES

- **Machine Learning & NLP -** Advanced Machine Learning, Natural Language Processing with Deep Learning (online), Automatic Speech Recognition[2], Foundations of Intelligent and Learning Agents[2], Artificial Intelligence and Machine Learning.
- **Electrical Engineering -** Adv. Concentration Inequalities[2], Information Theory, Optimization[2].
- **Computer Science -** Data Structures and Algorithms, Discrete Structure, Logic for CS, Digital Logic Design, Design and Analysis of Algorithms, Automata Theory, Compilers, Operating System, Database and Information Systems, Computer Networks, Computer Architecture.
- **Mathematics -** Multivariate Calculus, Linear Algebra, Differential Equations, Numerical Analysis.

TECHNICAL STRENGTHS

Strong	Python (with Tensorflow and Pytorch), C/C++, Bash
Familiar	Java, MATLAB, PHP, Android, VHDL, Arduino
Tools	Doxygen, Git, SVN, LaTeX

RESPONSIBILITIES

Academic Coordinator, *EnPoWER* 2017-18
Undergraduate Academic Council

· Assisted in the organisation of Summer Undergraduate Research Program involving 240+ students.
· Facilitated in the development of an online portal to centralize research projects in the institute.

EXTRA-CURRICULAR ACTIVITIES

- Played 'Tabla', an Indian percussion instrument for six years with various performances.
- Completed a year long course in badminton as a part of the National Sports Organization.
- Selected for 'Spark: Catch Them Young' program at Infosys, a 2 week training program in CS.

REFERENCES

Prof. Preethi Jyothi
IIT Bombay
webpage ⋄ email

Prof. Mohit Bansal
University of North Carolina, Chapel Hill
webpage ⋄ email

Prof. Ganesh Ramakrishnan
IIT Bombay
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Dr. Christopher Malon
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[2] Courses taken in Fall 2019